Predicting First-Passage Dynamics in Disordered Systems Exactly: Application to Sparse Networks

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Quantifying how spatial disorder affects the movement of a diffusing particle or agent is fundamental to target search studies. When diffusion occurs on a network, that is on a highly disordered environment, we lack the mathematical tools to calculate exactly the temporal characteristics of search processes, instead relying on estimates provided by stochastic simulations. To close this knowledge gap we devise a general methodology to represent analytically the movement and search dynamics of a diffusing random walk on sparse graphs. We show its utility by uncovering the existence of a bi-modality regime in the time-dependence of the first-passage probability to hit a target node in a small-world network. By identifying the network features that give rise to the bi-modal regime, we challenge long-held beliefs on how the statistics of the so-called direct, intermediate, and indirect trajectories influence the shape of the resulting first-passage and first-absorption probabilities and the interpretation of their mean values. Overall these findings show that temporal features in first-passage studies can be utilised to unearth novel transport paradigms in spatially heterogeneous environments.

Changes in the movement statistics of an agent as a result of interactions with a disordered environment is a ubiquitous phenomenon. Applications appear both in natural and engineered systems, ranging from the flow of substance through geological or porous substrates [1], the motion of charge carriers in composite battery architectures [2], and the slowing down of colloidal particles in glassy systems [3], to the migration of cells through the extracellular matrix [4] or the dispersal of animals in the landscape [5].

The theoretical physics literature devoted to quantify the effects of spatial heterogeneities on the transport of diffusive particles is vast and with a long history [6]. Some of the recent efforts to analyse these effects have focused on first-passage (FP) times of random walk search in disordered lattices [7–10]. While early work in this regard has focused on quantifying the mean first-passage time (MFPT) [11–13], it is now established that relying exclusively on the MFPT risks overestimating empirical FP times, questioning the use of a single numerical value to characterise search processes [14, 15]. To remedy this a formalism to extract the full FP probability for a disordered 1D lattice has been proposed [7], but it is only valid for nearest-neighbour jump processes, limiting its applicability to more general disordered structures.

To overcome these limitations, we present a general approach that allows to quantify the FP probability in arbitrary disordered lattices. Since networks structures are the most general example of a disordered environment, tools to quantify the FP statistics of a random walker over a network are highly sought after. In network science predicting the time to reach a given node for the first time already serves multiple purposes. From a theoretical perspective it allows to identify influential nodes [16] through generalized measures of closeness [17] and betweenness centralities [18], which both consider averages over all random trajectories rather than shortest paths. From the application point of view FP statistics have been used to identify spatial correlations in election results [19] as well as to de-noise [20] and segment images [21].

When a random walker is confined to a network, various approaches to study both the FP probability and its mean have been suggested. For arbitrary networks numerical techniques exist, e.g. the renewal analysis [22], the calculation of the adjacency matrix eigenspectrum [23, 24] or iterative procedures [25]. For specific graphs, a recent noteworthy technique is the methodology devised for dense networks [26] whereby one approximates the MFPT in a computationally efficient way using only the neighbourhood of the target location. This approach, however, becomes impractical in sparse networks, which is expected as spectral methods are known to break down in these cases [27].

One important class of sparse graphs, characterized by their high clustering and short path lengths, is the small-world network (SWN), which are constructed by rewiring regular ring lattices [28–32]. Numerous works studying the random walk statistics on SWNs have appeared [33–35], some of which play an important role in modelling the propagation of disease and rumours through a population [33, 36, 37]. In the context of search statistics, an approximate scaling law for the MFPT as a function of the disorder parameter was found through a fitting pro-

cedure to numerical data [38]. While such an approach gives good results when the number of nodes is large, it shows considerable deviation from the actual MFPT otherwise [22, 25, 39].

As many of the challenges to determine with high accuracy the first-passage dynamics relate to the lack of explicit mathematical representation of the diffusive dynamics on networks, we present a general analytical methodology to construct the occupation probability dynamics of a random walk moving randomly on a graph. As our interest is the dynamics on SWNs, we consider an arbitrary K-neighbour random walk on the ring lattice, and derive previously unknown expressions for its occupation probability and the MFPT to a single target.

Dynamics on the ring lattice — To study the random walk dynamics on a ring lattice, we start by describing it in an unbounded 1D domain. The Master equation for the occupation probability $P(n,t)$ of a $K = 2k$ neighbour lattice random walk is subject to

$$P(n,t+1) = \frac{1}{K}\sum_{r=1}^{k}\Big[P(n-r,t) + P(n+r,t)\Big], \quad (1)$$

which states that at each (discrete) time t, the walker may jump to site n from the k site(s) to the left or to the right of n. We have omitted the probability of staying at site n from the dynamics since we will consider the Watts-Strogatz SWN network, which typically does not have self-loops. If necessary, one may straightforwardly account for self loops by multiplying $q \in (0,1)$ to the term on the right hand side of Eq. (1) and by adding the term $(1-q)P(n,t)$.

To find the dynamics of the occupation probability $Q(n,t)$, where the shift from $P(n,t)$ to $Q(n,t)$ denotes that the random walker is confined to a finite lattice, we follow standard procedures. Namely, we consider a ring of N lattice sites, and Fourier and z-transform [40, 41] Eq. (1), before applying the method of images [40, 42] for an initially localised probability $Q(n,0) = \delta_{n,n_0}$. In such a case the so-called propagator solution $Q_{n_0}(n,t)$ of the Master equation, that is Eq. (1) supplemented by the periodic boundary condition $Q(n,t) = Q(n+N,t)$, is given by the generating function, $\widetilde{Q}_{n_0}(n,z) = \sum_{n=0}^{\infty} Q_{n_0}(n,t)z^t$ [43],

$$\widetilde{Q}^{(k)}_{n_0}(n,z) = \frac{1}{N}\sum_{\ell=0}^{N-1}\frac{\cos\left(\frac{2\pi\ell(n-n_0)}{N}\right)}{1-\frac{z}{k}\sin\left(\frac{k\ell\pi}{N}\right)\cos\left(\frac{(k+1)\ell\pi}{N}\right)\csc\left(\frac{\ell\pi}{N}\right)}, \quad (2)$$

where we have used the superscript (k) to make explicit the dependence on k (or K). The time dependent $Q^{(k)}_{n_0}(n,t)$ is trivially obtained from (2). Note that when $k \in \{1,2,3\}$ one may avoid the need to represent the propagator as a finite series by exploiting the connection between certain finite trigonometric series and Chebyshev polynomials [43].

With Eq. (2), we employ the reactive defect technique to extract the first-absorption probability generating function [24, 44] as

$$\widetilde{A}_{n_0}(n,z) = \rho\,\widetilde{Q}^{(k)}_{n_0}(n,z)/\left[1-\rho+\rho\,\widetilde{Q}^{(k)}_{n}(n,z)\right], \quad (3)$$

where $\rho \in (0,1]$ is the probability of the search being successful, that is the probability that the walker is absorbed whenever it is located at the target site n. From Eq. (3) one deduces the FP probability generating function in the limit $\rho = 1$, and the mean first-absorption time [24] $\mathcal{A}_{n_0\to n}(\rho) = \mathcal{F}_{n_0\to n} + \frac{1-\rho}{\rho}\mathcal{R}_n$, which is trivially related to the MFPT $\mathcal{F}_{n_0\to n}$ and the mean return time (MRT) to n, $\mathcal{R}_n$.

With the propagator in Eq. (2), the MFPT and MRT can be extracted algebraically from the probability generating function using renewal relations [41, 45]. For the MFPT we obtain

$$\mathcal{F}^{(k)}_{n_0\to n} = k\sum_{\ell=1}^{N-1}\frac{\left[1-\cos\left(\frac{2\pi\ell(n-n_0)}{N}\right)\right]\sin\left(\frac{\ell\pi}{N}\right)}{k\sin\left(\frac{\ell\pi}{N}\right)-\sin\left(\frac{k\ell\pi}{N}\right)\cos\left(\frac{(k+1)\ell\pi}{N}\right)}. \quad (4)$$

Note that if self loops are present, i.e. $q \neq 1$, Eq. (4) is modified by a q^{-1} multiplication factor. The MRT is found as $\mathcal{R}_n = N$, regardless of k, which is expected since the general MRT on a network may be written as $\mathcal{R}_n = \mathcal{E}/\chi_n$, where $\mathcal{E}$ is the number of edges in the network and χ_n is the degree (co-ordination number) of node n [22]. Two well-known limits of Eq. (4) can be recovered. The $k = 1$ case reduces to [42] $\mathcal{F}^{(1)}_{n_0\to n} = (N - |n-n_0|)\,|n-n_0|$. For the opposite limit, the fully connected graph, we take $k = \frac{N-1}{2}$ ($N \in 2\mathbb{N}+1$) and we obtain [26, 38] $\mathcal{F}^{\left(\frac{N-1}{2}\right)}_{n_0\to n} = (1-\delta_{n,n_0})(N-1)$ (see [43] for the derivation of the same expression when N is even).

Dynamics on the network — To build the SWN, we follow the Watts-Strogatz prescription [28], whereby a regular ring lattice is transformed through the introduction of random re-wirings governed by some probability p [47]. As p is increased, the structure of the graph becomes more disordered, approaching a random network as $p \to 1$.

Each of the rewired links in the SWN represents a spatial *defect* of the inert type (probability preserving) of an otherwise homogenous ring lattice. The number of required defects is dependent on p, N and k. In particular, the average number of defects for a given set of parameters is

$$M = (K+1)pN, \quad (5)$$

which arises as each SWN has, on average, pN re-wirings [28] and each re-wiring requires $K+1$ defects. This ensures that the probability of taking each outgoing edge from site n is maintained as $1/\chi_n$, i.e., upon cutting or adding a link at n, the coordination number of that node

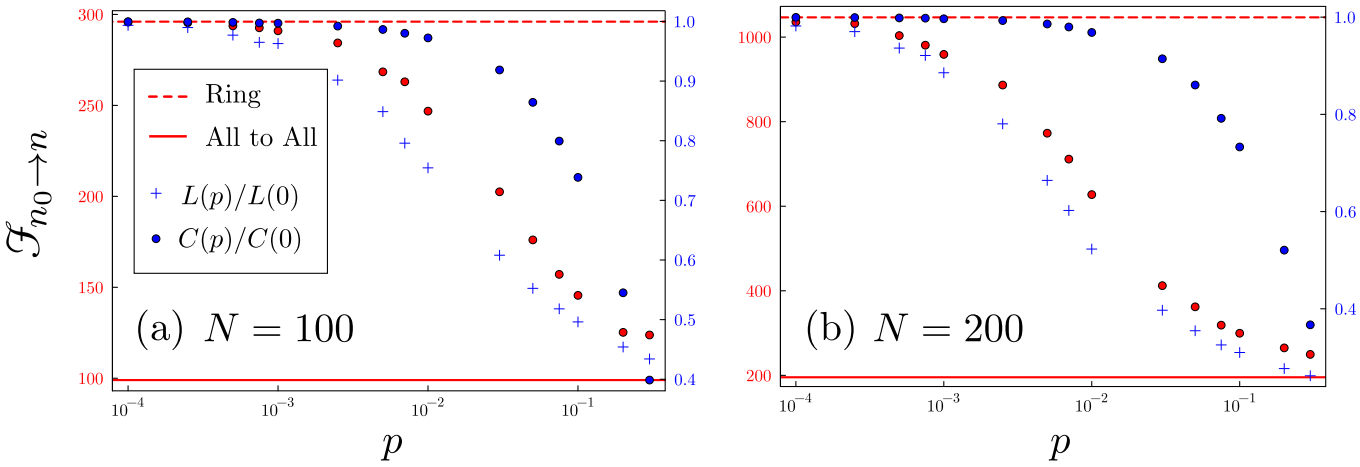

FIG. 1. A comparison between the MFPT (red axis on the left) and the graph properties of the characteristic path length, $L(p)$, and clustering coefficient, $C(p)$, (blue axis on the right) on small-world networks with $K = 10$. On the right axis we follow ref. [28] and normalise the coefficients by the appropriate coefficient for the ring lattice ($C(0)$ or $L(0)$). For all MFPT calculations, we take $n_0 = 1$ and $n = N/2$, i.e., the longest path distance on the ring lattice. We also display the result on the $K = 10$ ring graph (dashed line) and the all-to-all graph (solid line). Each point and cross in both plots represents an ensemble average of 50 realisations of the disorder for each p. We determine the characteristic path length by averaging the shortest path, found via Dijkstra's algorithm [46], over all start and end nodes.

has changed, meaning every other edge also requires its probability to be altered. Hence, the procedure is particularly useful when $K + 1 < 1/p$ i.e., for sparse networks (low k) with high clustering coefficients (small p), since the size of the defect matrices required to find the occupation probability exactly (explained below) is smaller than $N \times N$.

In the presence of M inert defects the Master equation governing the lattice random walk probability can be expressed by directly altering the transition probabilities via [10]

$$S(n, t+1) = \sum_{n'} \mathbb{B}_{nn'} S(n', t) + \sum_{m=1}^{M} (\delta_{n,u_m} - \delta_{n,v_m}) \left[\eta_{v_m,u_m} S(u_m, t) - \eta_{u_m,v_m} S(v_m, t) \right], \tag{6}$$

where $\mathbb{B}_{nn'}$ represents the defect-free transition probability from n' to n, $\delta_{a,b}$ is a Kronecker delta, and $\eta_{x,y}$ quantifies the alteration of the probability of jumping from y to x. In other words $S(n, t+1) = \sum_{n'} \mathbb{B}_{nn'} S(n', t)$ represents the defect-free Master equation for the lattice walker on the ring lattice, namely Eq. (1).

From Eq. (6), one sees that each re-wiring to build the SWN from the ring lattice requires four pieces of information regarding the inert defects, namely $(u, v, \eta_{v,u}, \eta_{u,v})$, which is conveniently represented algorithmically as a four-tuple. To extract the set of four-tuples note that each $\mathbb{B}$ is a symmetric circulant matrix with $A_{ij} = 1/K$ if $0 < |i - j| \bmod N \leq k$, and $A_{ij} = 0$ otherwise. Similarly, one may define a transition matrix containing the jump probabilities on the network as $\mathbb{A}$, which is easily extracted via Graph analysis libraries e.g., Graphs.jl [48] or NetworkX [49]. Using this formulation we find the modifications, or defect parameters as the non-zero elements of $\mathbb{X} = \mathbb{B} - \mathbb{A}$ where each four-tuple is $(i, j, \mathbb{X}_{j,i}, \mathbb{X}_{i,j})$.

With the dynamics on the ring lattice and the set of defects established, the formalism of ref. [10] may now be used to find analytically the occupation probability and the MFPT, respectively. To make conspicuous the change from the ring lattice to the network we use $\mathcal{F}_{n_0 \to n}$ instead of $\mathcal{F}_{n_0 \to n}$ to denote the MFPT on the SWN. The propagator generating function of Eq. (6) and $\mathcal{F}_{n_0 \to n}$ are expressed analytically in terms of determinants of $M \times M$ matrices, whose exact analytic dependence are given explicitly in ref. [43], and comprise solely expressions pertaining to the ring lattice, respectively Eqs. (2) and (4), and the various η parameters.

In a SWN, with rewired links that massively cut the typical separation between two vertices, we expect that a portion of the lattice walk trajectories no longer traverse the entire outer ring to reach a given target [38]. This effect is visible in Fig. 1, where we plot the MFPT as a function of the re-wiring parameter p. As a quantitative comparison we show explicitly that a random walk average search time lies between two limits, the MFPT on the ring lattice and the one on a fully connected graph. Those two extreme values are bridged through a rapid decline occurring at the same p-threshold for which the characteristic path length drops suddenly, i.e. at the onset of the small-world property.

As the MFPT hides much of the nuances of the search behaviour, in Fig. 2 we explore the FP distribution for

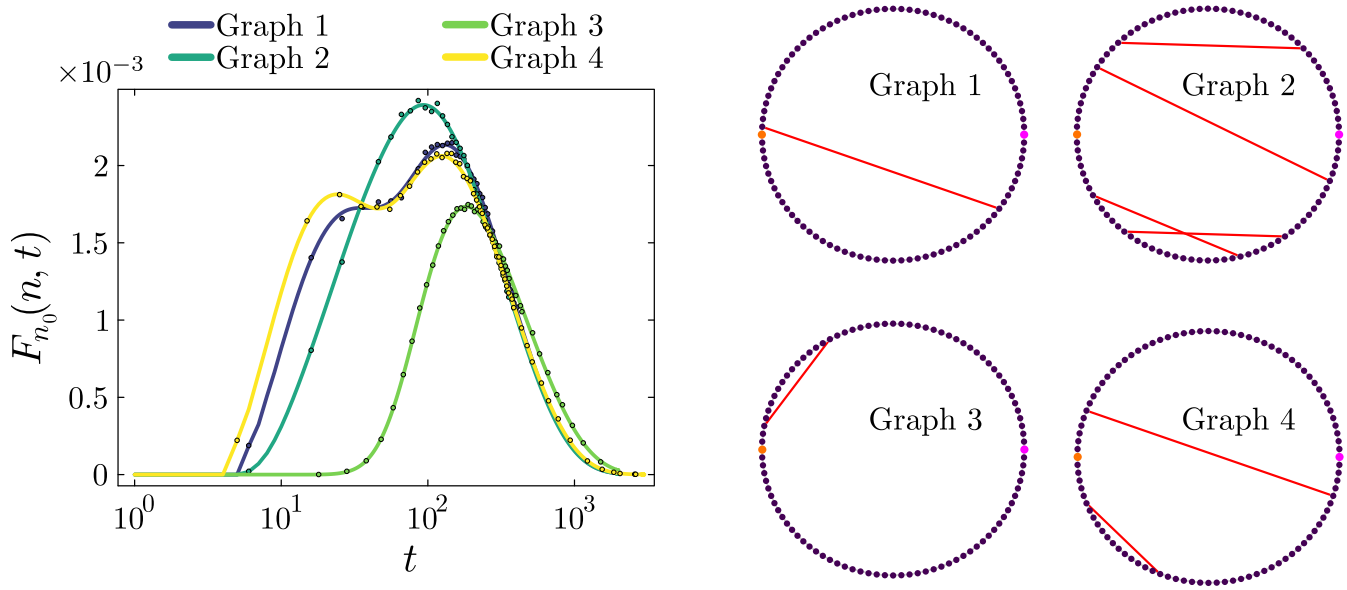

FIG. 2. A comparison of the FP distributions for four independent realisations of the SWN constructed from $K = 6$ ring lattice. On the left each analytical curve (solid line) is verified against stochastic simulations (dots), while the schematics of each of the network structure is displayed on the right. For each realisation the network is generated with $p = 0.15$ and $N = 100$, and the initial ($n_0 = 1$) and target ($n = 50$) node are shown as slightly larger than the other nodes and depicted, respectively, in orange and pink colour. For ease of visualisation and to give prominence to the different types of disorder, we have removed the K neighbour connections from the network schematics and coloured the long range connections red.

four realisations of the SWN with the same number of nodes and edges, the same rewiring parameter, and the same initial and target location. The variation in transport characteristics between the four SWNs manifests itself in the qualitative differences of the FP distributions. This is particularly evident in Graph 4 where the FP distribution is bi-modal. In that case those walkers that traverse the shortcut early on and then reach the target characterise the *direct* trajectories [15] and give rise to the first mode of the FP probability. While the tail of the FP distribution is controlled by the so-called *indirect* trajectories with long excursion across the entire domain, there exists a temporal regime with trajectories, referred to as *intermediate*, that make initial excursions away from the target, before proceeding directly to it [50]. We posit that a pronounced timescale separation between direct and intermediate trajectories and the presence of strong spatial heterogeneities that enhance the latter is what leads to a regime of bi-modality in the FP probability.

To confirm the above assertion, we artificially create a network with a structure similar to Graph 4 of Fig. 2. We do that by taking a ring network with a given K and introducing a 'helpful' connection by placing one short-cut between nodes $n + 1$ and $n_0 + 5$, where n is the target node, thus having $M = K + 1$ defects in Eq. (6). In this way, by engineering a connection that links the target and initial node with a small number of steps, we control the direct trajectories and thus the timescale of the first mode. Keeping this local network structure, we study how the number of nodes, N, connectivity K, and the absorption parameter at the target, ρ, affect the intermediate trajectories and, in turn, the appearance of a second mode in the FP probability, as displayed in Fig. 3(c) for certain parameter choice (see also Fig. S2 in [43]).

In Figs. 3(a) and 3(b) we identify with coloured cells the regime of bi-modality when a walker is absorbed at n starting from n_0. The black areas in the panels correspond to when the FP probability is unimodal, where, the FP mode at the top being always smaller, for the same value of ρ, than the one at the bottom. The area in between corresponds to when both modes are present and points to two features for their simultaneous appearance. On one hand the ratio N/K needs to be large enough (above the bottom black region) such that most direct FP trajectories, which give rise to the first mode, $\mathfrak{F}^{(1)}_{n_0}(n)$, are absorbed before those that traverse the outer ring, the latter ones giving rise to the second mode, $\mathfrak{F}^{(2)}_{n_0}(n)$. On the other hand N/K cannot be too big (below the top black region) to avoid large variation in search times of the outer intermediate trajectories that would make the second mode disappear. See Fig. S2 in ref. [43] for a visual comparison of these distribution types.

As an imperfect absorption ($\rho < 1$) has the effect of

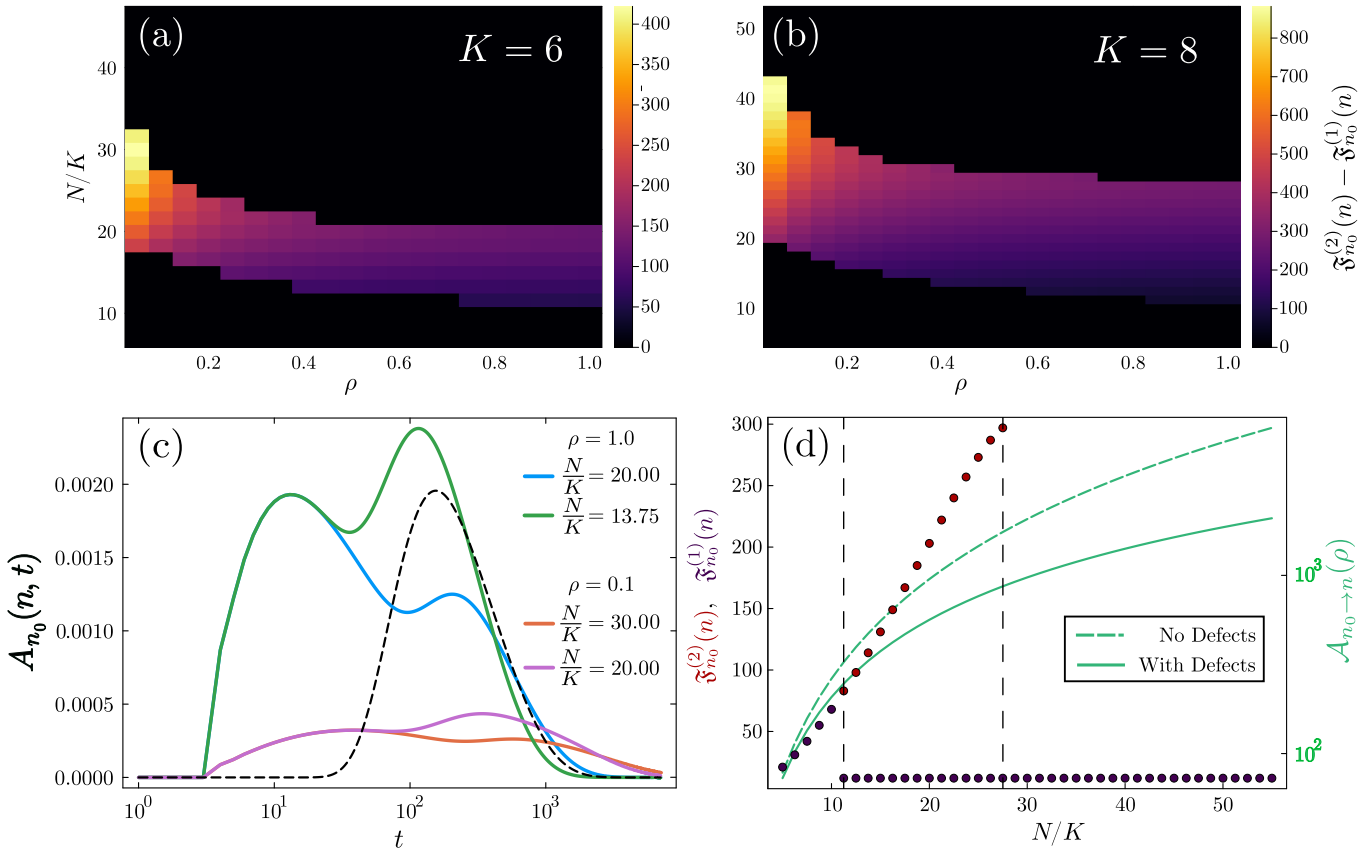

FIG. 3. Bi-modality in the first-absorption distribution from $n_0 = 1$ to a target at $n = N/2$ with N even on an artificially generated small world network (see text), obtained by replacing $\widetilde{Q}^{(k)}_{n_0}(n,z)$ with the solution $\widetilde{S}_{n_0}(n,z)$ of (6) in Eq. (3) and inverting the resulting generating function to time. In panels (a) and (b) non-black coloured cells indicate the presence of two modes with the colour intensity representing the amount of time between the locations of the two modes ($\mathfrak{F}^{(i)}_{n_0}(n)$ denotes the time of the ith mode). Panel (c) shows examples of the entire distribution for different network parameters in the bi-modal regime, with the defect-free case (no re-wiring) when $\rho = 1$ as a dashed curve. Panel (d) compares the dependence of the MFPT, that is $\mathcal{A}_{n_0 \to n}(\rho = 1)$, and the first and second mode, which are both present in between the dashed vertical lines, as a function of the network size N ($K = 8$). Across all plots we identify the presence of a mode as a time t when $F_{n_0}(n, t-1) < F_{n_0}(n,t)$ and $F_{n_0}(n,t) > F_{n_0}(n, t+1)$. To avoid potential misclassifications we demand that a peak satisfies the inequality $F_{n_0}(n,t) > 10^{-7}$ and, if a second mode exists, it must be at least 1% of the height of the highest one.

extending the length of all FP trajectories, the two FP modes get reduced in magnitude and become broader the smaller the value of ρ. For a given ratio N/K, by taking progressively smaller values of ρ, the trajectories taking the shortcut wander for increasingly longer time before being absorbed at the target. As the absorption timescale for such trajectories acquire large variability and become comparable to some of the movement paths that go around the outer ring, for a set N/K, below a threshold value of ρ there is no second mode. In such scenario, without changing ρ, bi-modality can be restored by increasing the size of the network, which allows to bring back the temporal separation between direct and intermediate trajectories. In fact, as larger K values increase variability in FP times, one may still observe two modes but needs to have a network with a larger number of nodes, which in turn makes absorption times for the outer trajectories take longer. These are the reasons why, in comparing panels (a) and (b), we observe an upward shift, a broadening of the bi-modality region and an increase in timescale separation between $\mathfrak{F}^{(2)}_{n_0}(n)$ and $\mathfrak{F}^{(1)}_{n_0}(n)$.

Past studies of heterogeneity-controlled kinetics, which uncovered the existence of a third timescale associated with intermediate trajectories, have concluded that the MFPT, while being dominated by indirect trajectories, is completely specified by the statistics of direct trajectories [50]. To bring evidence that this statement is not universally valid, in Fig.3(d) we show that in the bi-modal region the MFPT does increase as the network size increases, but without any variation in $\mathfrak{F}^{(1)}_{n_0}(n)$, that is leaving the statistics of direct trajectories unchanged (for $\rho \neq 1$ see [43]). For sufficiently small N/K values we also observe that $\mathfrak{F}^{(2)}_{n_0}(n)$ becomes the largest characteris-

tic timescales in the system, even larger than the MFPT. Given the non-homogeneous nature of our small world network, as compared to the radially symmetric arrangement studied in ref. [50], we deduce that in strongly heterogeneous environments, only two typical FP timescales may appear, the early one associated with direct trajectories, and the later one associated with intermediate trajectories, and that the MFPT is not only independent of the statistics of direct trajectories, but loses its meaning as representing the average time for the indirect trajectories to be absorbed at the target.

Discussion — In this Letter we provide a general framework to study analytically the first-passage probability on network structures devising a procedure to exploit the inert defect technique [10]. For our application to SWNs we derive previously unknown expressions for the dynamics of the probability of the k nearest-neighbour random walk. In doing so, we show that the spatio-temporal search dynamics of a random walker on a SWN is highly sensitive to the network structure. We uncover, for the first time in symmetric and diffusive systems, bi-modal first-absorption distributions, which are driven by the ratio of the network size and the number of nearest neighbours in the ring lattice. We also provide evidence that the mode at short times is dependent on the local network structure around the initial condition and target location, that the later mode is dependent on the global network structure, and that in the bi-modal regime the MFPT is not affected by the statistics of the direct trajectories and cannot be interpreted as the characteristic timescale of indirect trajectories.

The strength of our approach lies in its general applicability. As input it requires mathematical knowledge of the dynamics of the occupation probability in a topology that is not too dissimilar from the specific problem at hand. Such scenario occurs in any graph that displays structural similarities to a strongly regular graph [51] or to some other specific graph over which the dynamics of the random walk occupation probability is known analytically. In these situations the entire time-dependent probability of the walk as well of the first-passage probability can be extracted without recurring to stochastic simulations. While the methodology becomes computationally onerous if the topology is very different from that of the associated defect-free case, that is when the number of defects become large, it is however valid irrespective of the strength of the defect. It is thus an effective approach when a walker interacts strongly with the disorder in the environment and ideally suited to link temporal features to spatial heterogeneity in FP processes.

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